Hugo Marquerie 17/10/2025

Teo metrica lp 0 1

Teorema 1. *Sea* (X, Σ, μ) *un espacio de medida y* $0 < p < 1$,

$$\implies \forall f, g \in L^p(X, \Sigma, \mu) : d_p(f, g) := \int_X |f - g|^p \,\mathrm{d}\mu$$

define una métrica en $L^p(X, \Sigma, \mu)$.

***Demostración*:** Solo falta probar la desigualdad triangular. Por el `lem-desigualdad-p-0-1/`**??**, tenemos que $\forall a, b \in \mathbb{R}_+ : (a + b)^p \leqslant a^p + b^p$. Sean $f, g, h \in L^p$, entonces

$$\begin{aligned} d_p(f, h) = \int_X |f - h|^p \,\mathrm{d}\mu &\overset{\triangle}{\overset{\downarrow}{\leqslant}} \int_X (|f - g| + |g - h|)^p \,\mathrm{d}\mu \\ &\leqslant \int_X |f - g|^p \,\mathrm{d}\mu + \int_X |g - h|^p \,\mathrm{d}\mu = d_p(f, g) + d_p(g, h). \end{aligned}$$

■

Temporary page!

LaTeX was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it.

If you rerun the document (without altering it) this surplus page will go away, because LaTeX now knows how many pages to expect for this document.